「Orthogonal Projection - Revised.

Let V be an inn. prod space, (not necessary finite dimensional) and $W \leq V$ be a subspace.
For any vector $\beta \in V$, if $\alpha \in W$ satisfies that:

$$\|\beta - \alpha\| \leq \|\beta - \gamma\| \quad \text{for all } \gamma \in W \quad \cdots (*)$$

Then, the α is called best approximation to β by W, simply b. app.

「Thm. $\alpha \in W$ is b.app to $\beta \in V$ by W iff for all $\mu \in W$, $\langle \beta - \alpha, \mu \rangle = 0$.

Proof. For fixed $\alpha \in W$ and $\beta \in V$, for any $\gamma \in W$,

$$\|\beta - \gamma\|^2 = \|\beta - \alpha + \alpha - \gamma\|^2 = \|\beta - \alpha\|^2 + \|\alpha - \gamma\|^2 + 2 \cdot \mathrm{Re}\langle \beta - \alpha, \alpha - \gamma \rangle$$

Therefore, the condition $(*)$ holds if and only if

$$0 \leq \|\alpha - \gamma\|^2 + 2 \cdot \mathrm{Re}\langle \beta - \alpha, \alpha - \gamma \rangle \quad \text{for all } \gamma \in W.$$

Now take $\gamma = \alpha + \dfrac{\langle \beta - \alpha, \alpha - \mu \rangle}{\|\alpha - \mu\|^2} \cdot (\alpha - \mu)$ for any $\mu \in W \setminus \{\alpha\}$.

Then, $$0 \leq \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} - 2 \cdot \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} = -\frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2}$$

And the inequality holds iff $\langle \beta - \alpha, \alpha - \mu \rangle = 0$, and all nonzero vectors in W can be represented by $\alpha - \mu$, thus proved.

Conversely, if $\beta - \alpha$ is orthogonal to every vector in W, then $\mathrm{Re}\langle \beta - \alpha, \alpha - \gamma \rangle = 0$, thus $\|\beta - \gamma\|^2 - \|\beta - \alpha\|^2 = \|\alpha - \gamma\|^2 \geq 0$. ☒

「Thm. If b. app to $\beta \in V$ by W exists, then it is unique.

Proof. Suppose that $\alpha_1, \alpha_2 \in W$ satisfies for all $\gamma \in W$,

$$\langle \beta - \alpha_1, \gamma \rangle = \langle \beta - \alpha_2, \gamma \rangle = 0.$$

Then, take $\gamma = \alpha_1 - \alpha_2$, so $\langle \alpha_1 - \alpha_2, \alpha_1 - \alpha_2 \rangle = 0$. ☒

「Def. For all $\beta \in V$, there is b.app $\alpha \in W$, then the map

$$P : V \to V : \beta \mapsto \alpha$$

is called an orthogonal projection on W.

Thm. Let W be a fin. dim subspace of inn. prod space V. Then, the orthogonal projection $P: V \to V$ exists. Furthermore, P is idempotent linear operator with $\operatorname{Im} P = W$, $\ker P = W^{\perp}$.

Proof. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for W. Then, a map

$$P : V \to V ; \beta \mapsto \sum_{i=1}^{n} \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$$

1). It maps β to orthogonal projection by W.

For each $1 \le j \le n$,

$$\left\langle \beta - \sum_{i=1}^{n} \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i , \alpha_j \right\rangle = \langle \beta, \alpha_j \rangle - \langle \beta, \alpha_j \rangle = 0$$

So, $\beta - P(\beta) \in W^{\perp}$ for all $\beta \in V$. (And, such a vector is unique, thus well-defined)

2). It is linear.

Let $\beta, \gamma \in V$ and $c \in F$ be given. Then,

$$P(c\beta + \gamma) = \sum \frac{\langle c\beta + \gamma, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i = c \cdot \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i + \sum \frac{\langle \gamma, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i = cP(\beta) + P(\gamma)$$

3). It is idempotent.

Given $\beta \in V$, $P(\beta) \in W$. So, $P^2(\beta) = P(\beta)$.

4). $\operatorname{Im} P = W$, $\ker P = W^{\perp}$.

$\operatorname{Im} P \subseteq W$ is clear, and for any $\gamma \in W$, $P(\gamma) = \gamma \in \operatorname{Im} P$. And,

$$\beta \in \ker P \iff P(\beta) = 0 \iff \langle \beta, \alpha_i \rangle = 0 \text{ for all } 1 \le i \le n \iff \beta \in W^{\perp}.$$

Note: let $E : V \to V$ be a projection. In general, E can not be determined by it's image.

For example, let $V = \mathbb{R}^3$, $W_1 = \operatorname{span}\{(1,0,0)\}$, $W_2 = \operatorname{span}\{(0,1,0), (0,0,1)\}$, $W_3 = \operatorname{span}\{(0,1,0), (1,1,1)\}$

Then, $V = W_1 \oplus W_2 = W_1 \oplus W_3$, but $W_2 \neq W_3$. Moreover,

$E : V \to V : (x_1, x_2, x_3) \mapsto (x_1, 0, 0)$, $E' : V \to V ; (x_1, x_2, x_3) \mapsto (x_1 - x_3, 0, 0)$

Both E and E' are projections with $\operatorname{Im} E = \operatorname{Im} E' = \operatorname{span}\{(1,0,0)\}$ but $E = E'$, $\ker E = W_2$ and $\ker E' = W_3$.

Corollary. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthogonal set of non-zero vectors in inn. prod space V. Then, for all $\beta \in V$, $\displaystyle\sum_{i=1}^{n} \frac{|\langle \beta, \alpha_i \rangle|^2}{\|\alpha_i\|^2} \le \|\beta\|^2$; the equality holds iff $\displaystyle\beta = \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$.

Proof. Take $\gamma = \sum [\langle \beta, \alpha_i \rangle / \|\alpha_i\|] \cdot \alpha_i$. Then, $\langle \beta - \gamma, \gamma \rangle = 0$. So,

$$\|\beta\|^2 = \|\gamma\|^2 + \|\beta - \gamma\|^2 \ge \|\gamma\|^2.$$